StaleCheck: find the stale claims in an LLM answer

Paste an answer. Get every time-sensitive claim re-checked against today's sources.

Each claim: supported, potentially changed or unverifiable, with a 0-100 freshness score.

Every verdict rests on a verbatim quote and a source link. Edits are suggested, never applied.

One integrated project: Bifrost for every model call, Tavily for live research.

The problem: confident answers go out of date

Prices, versions, support windows and laws change after a model's training data ends.

The answer still sounds certain, and never says when its facts were true.

A reader cannot tell which sentence went stale.

Checking every claim by hand does not scale.

How a check runs

01

Bifrost extracts time-sensitive claims as exact spans of the answer

→

02

You approve the exact Tavily queries and domains

→

03

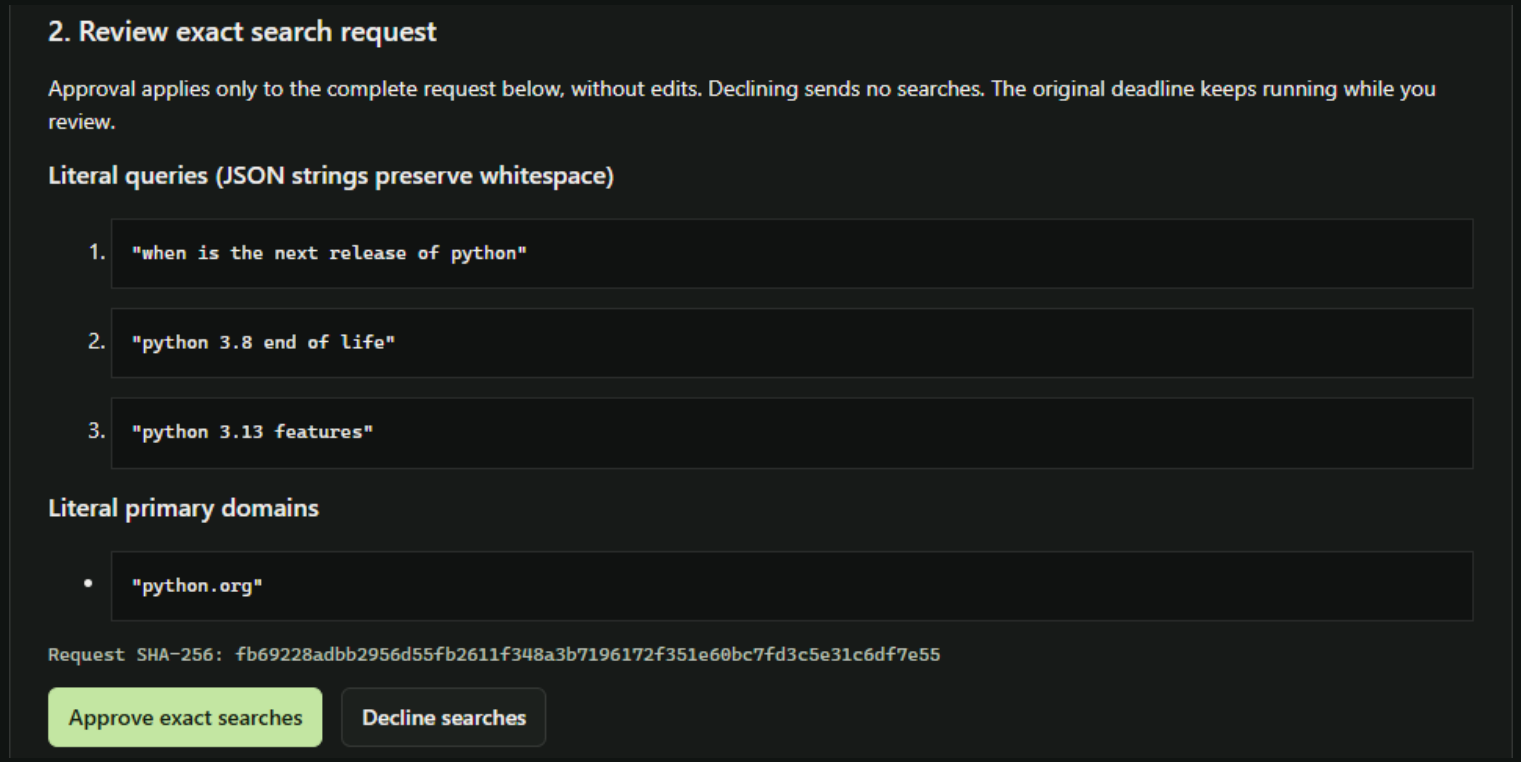
Tavily searches reviewed domains; Bifrost reads the excerpts

→

04

Host checks quotes verbatim, scores freshness, writes the report

Step 2: nothing is searched until you approve it



Live run: the exact Tavily queries the model proposed, for python.org only. Declining sends nothing.

Live result: real claims, real sources, real verdicts

At a glance

One row per selected claim. Freshness (0-100) is the recency of the quoted evidence, not proof that the claim is current.

Claim	Text	Category	Verdict	Freshness
claim-1	Python 3.12 is the newest stable release of Python.	version	POTENTIALLY_CHANGED	93
claim-2	Python 3.8 reached end of life in October 2024	version	UNVERIFIABLE	0
claim-3	The experimental free-threaded build, which removes the GIL, ships with Python 3.13.	version	POTENTIALLY_CHANGED	8

Real Bifrost 2.4.0 gateway, local Qwen2.5-3B model, real Tavily API (2026-11-02).

Evidence, not opinion

claim-1: POTENTIALLY_CHANGED

MODEL_CONTRADICTION_WITH_VERBATIM_QUOTE. Answer span [0, 45).

Original excerpt and proposed edit (never auto-applied)

Original	Model-generated proposal
Acme Cloud's Starter plan costs $19 per month	Acme Cloud's Starter plan costs $24 per month and includes 5 seats.

Category (model-labelled): pricing. Freshness of the quoted evidence: current (score 97/100, newest quoted source 2026-10-20, 12 day(s) before the assessment date). Evidence recency, not proof that the claim is current.

Demo data: the original sentence beside a model-proposed edit, labelled and never auto-applied.

How Maxim Bifrost and Tavily are used

Bifrost: the single gateway for claim extraction and assessment, via /v1/chat/completions with a virtual key.

Bifrost: requested fallbacks, plus the routing and token usage it reports, recorded for every call.

Tavily: one Search per approved query, restricted to reviewed domains.

Tavily: its excerpts are the only text a quote can come from; its dates drive the freshness score.

What the live run proved

End to end on real services: Bifrost 2.4.0 and the Tavily API, about 80 s and 3 credits per answer.

The guard works: a model verdict without a contradicting quote became unverifiable.

Failures stay safe: a gateway timeout was recorded, not retried, and the report still verified.

Still open: a hosted model and an accuracy evaluation. One answer is not a benchmark.

What makes it different

Model output is untrusted: a verdict changes only on a quote found verbatim in a source.

Two consent steps, bounded attempts and time, no automatic retries.

Runs are hashed and replayable; tampering is detected on verification.

Honest limits: unverifiable is shown, never guessed.

Try it

Offline demo, no keys: python -m stalecheck stale-check --demo-v4 --output runs/demo

Web app: python -m stalecheck serve-assessment --output runs/session

Live: your Bifrost virtual key and Tavily key, plus --live --confirm-outbound

Stack: Python 3.12 stdlib, React + TypeScript, pytest and Vitest, GitHub Actions CI